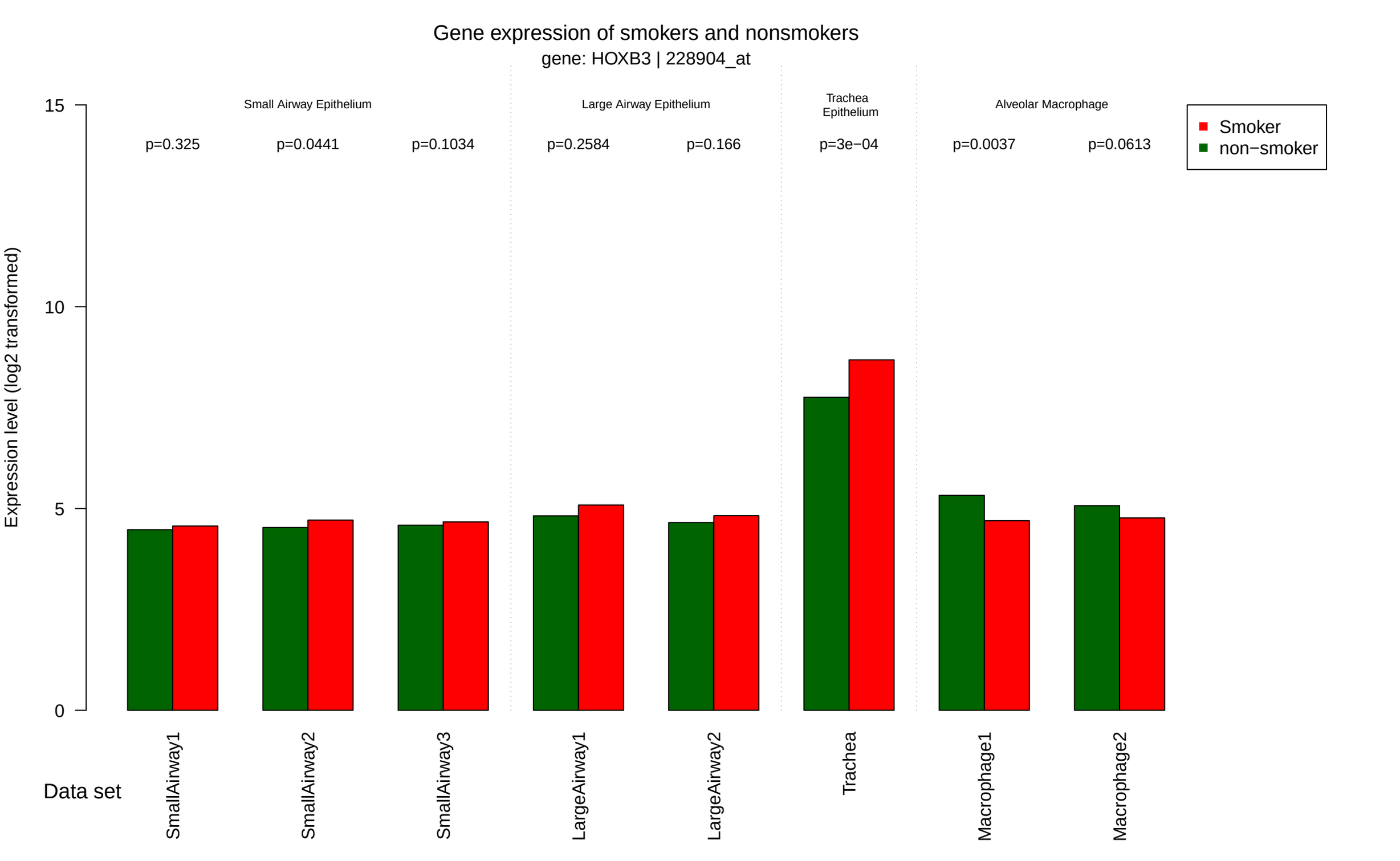
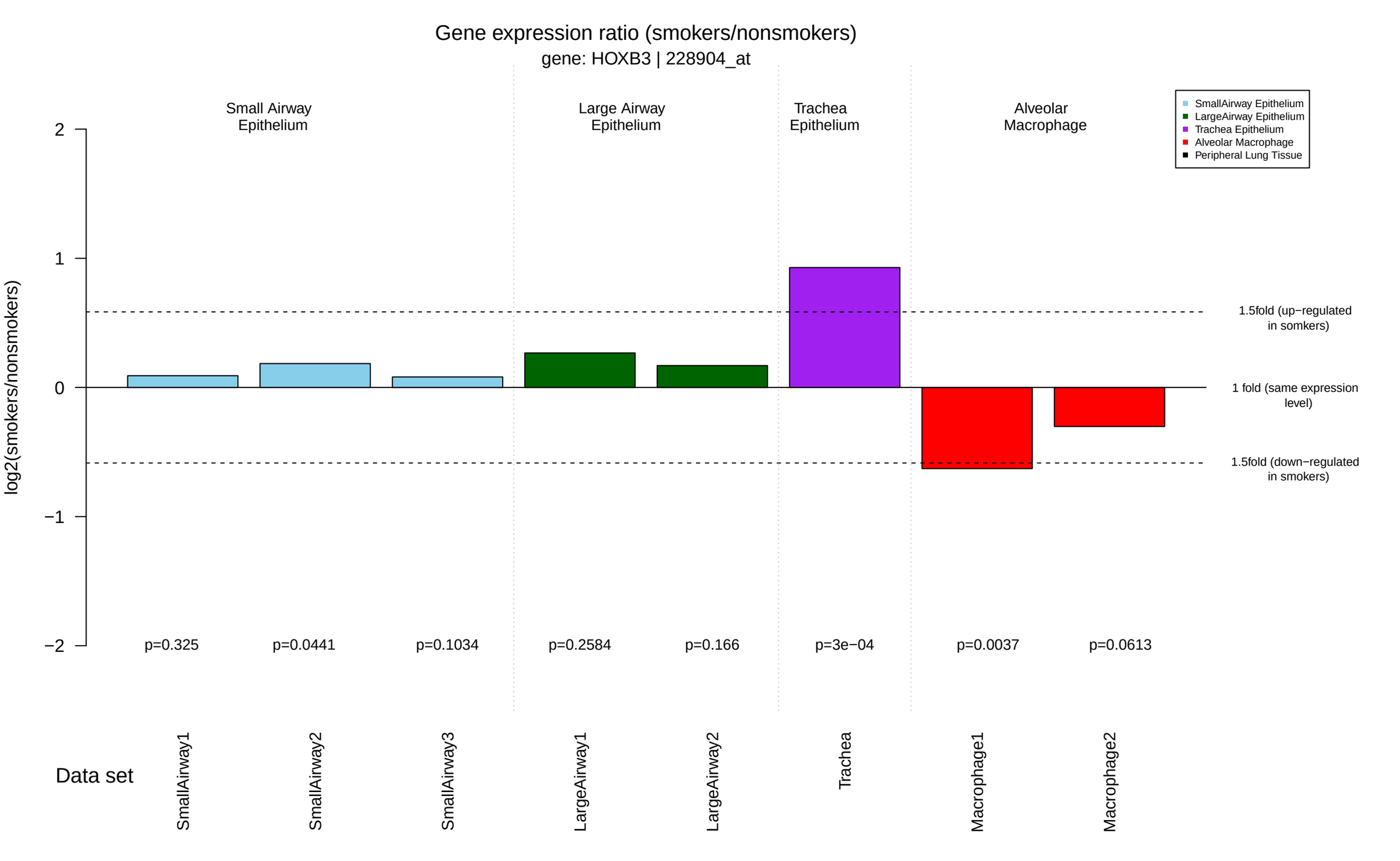
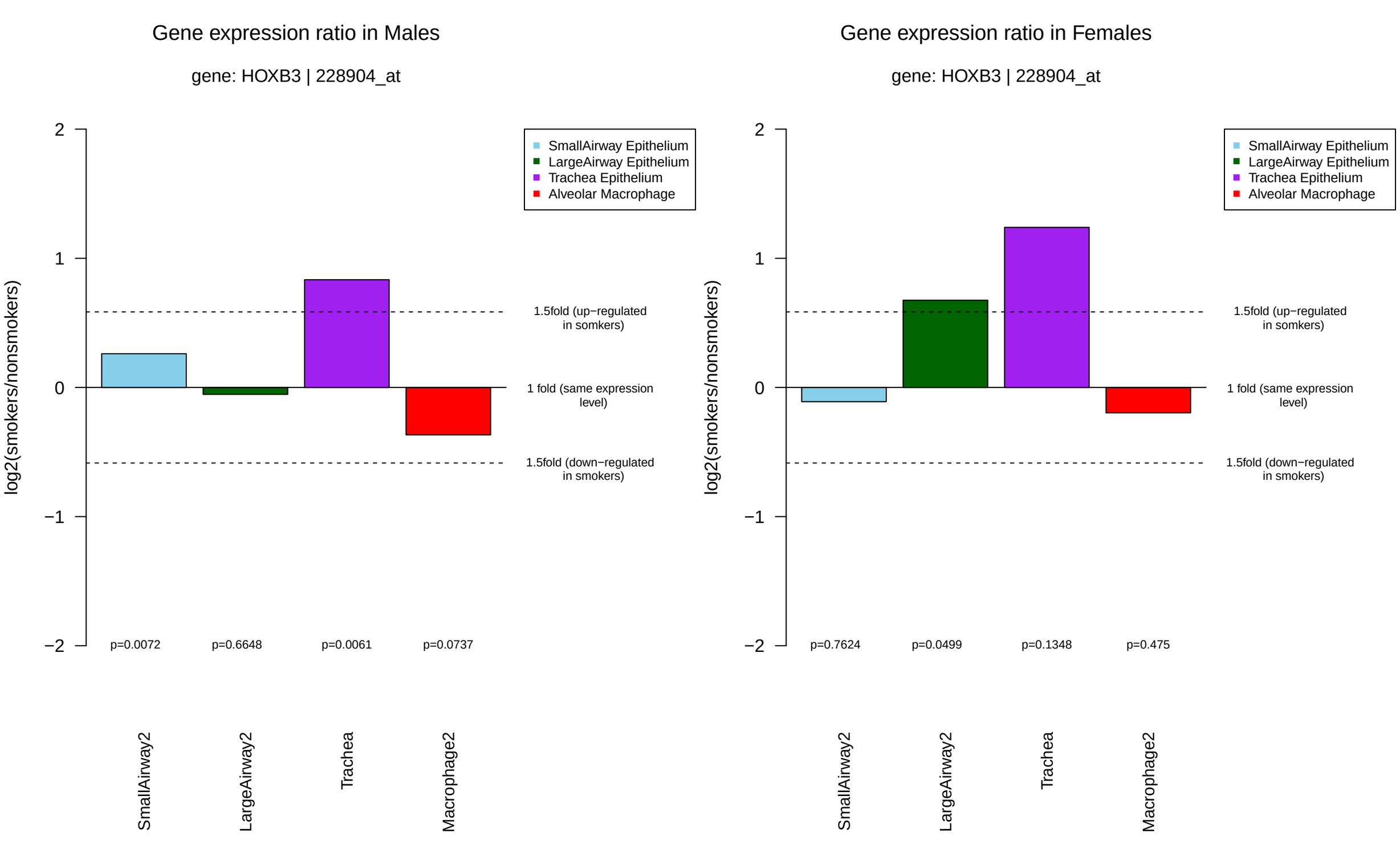
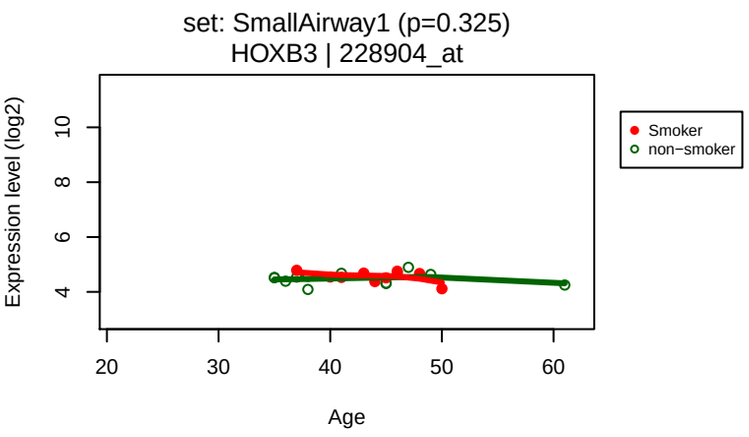
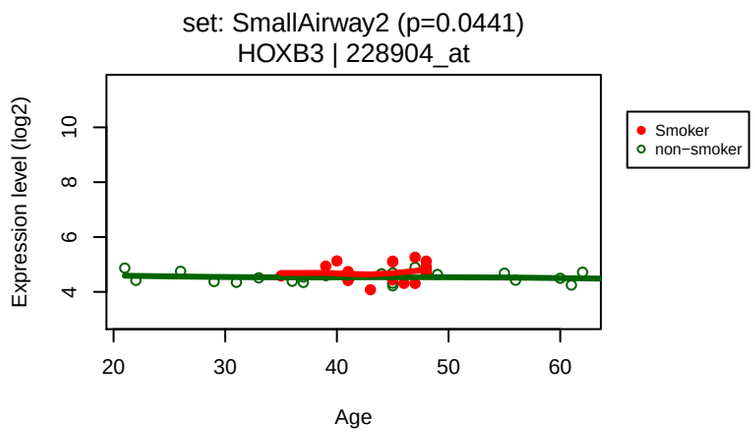
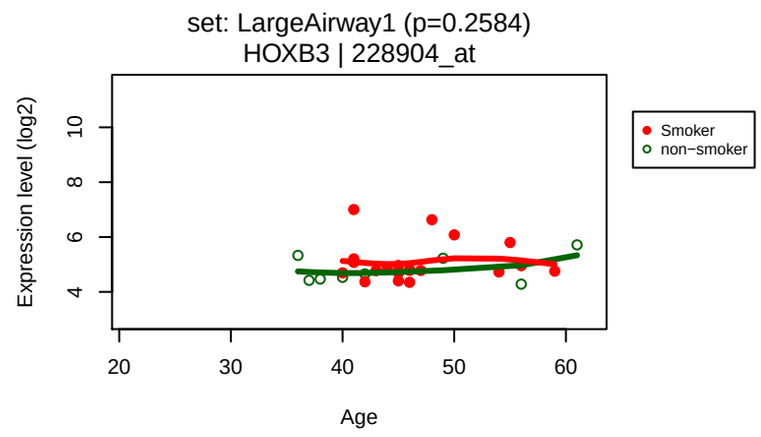

Gene expression of smokers and nonsmokers
gene: HOXB3 | 228904_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.325
p=0.0441
p=0.1034
p=0.2584
p=0.166
p=3e−04
p=0.0037
p=0.0613
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

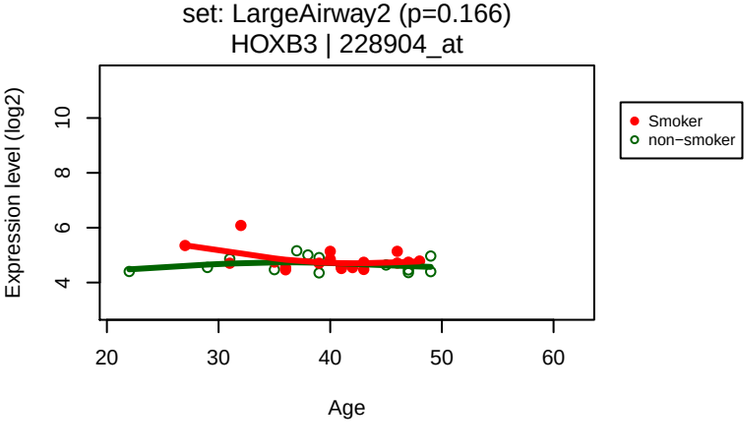

Gene expression ratio (smokers/nonsmokers)
gene: HOXB3 | 228904_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.325
p=0.0441
p=0.1034
p=0.2584
p=0.166
p=3e−04
p=0.0037
p=0.0613
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

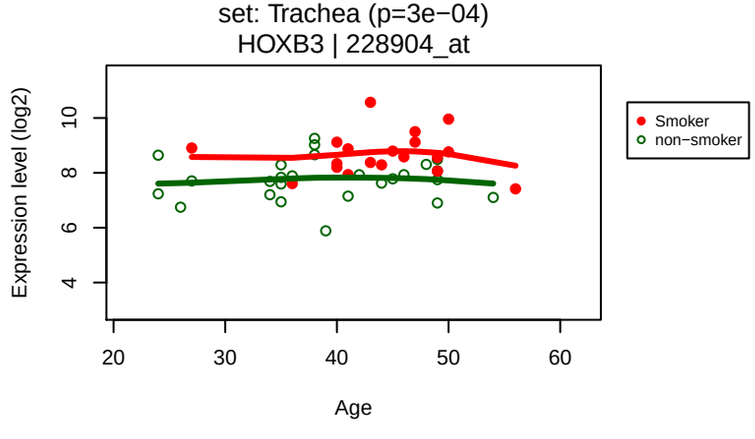

Gene expression ratio in Males
gene: HOXB3 | 228904_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0072
p=0.6648
p=0.0061
p=0.0737
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB3 | 228904_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7624
p=0.0499
p=0.1348
p=0.475
SmallAirway2
LargeAirway2
Trachea
Macrophage2

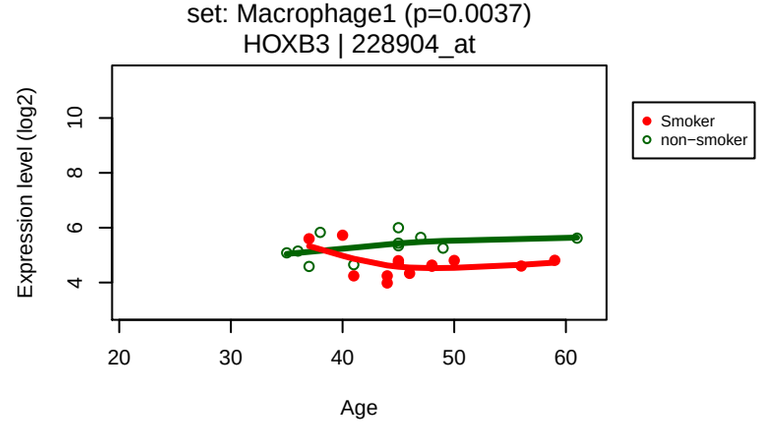

set: SmallAirway1 (p=0.325)
HOXB3 | 228904_at
Expression level (log2)
Age
Smoker
non−smoker

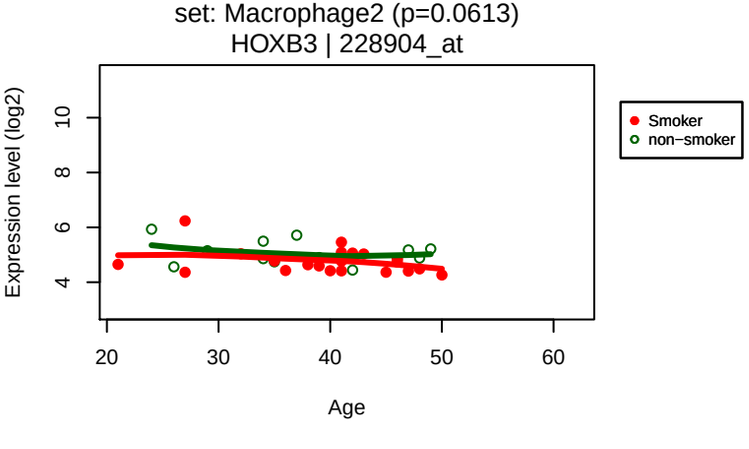

set: SmallAirway2 (p=0.0441)
HOXB3 | 228904_at
Expression level (log2)
Age
Smoker
non−smoker

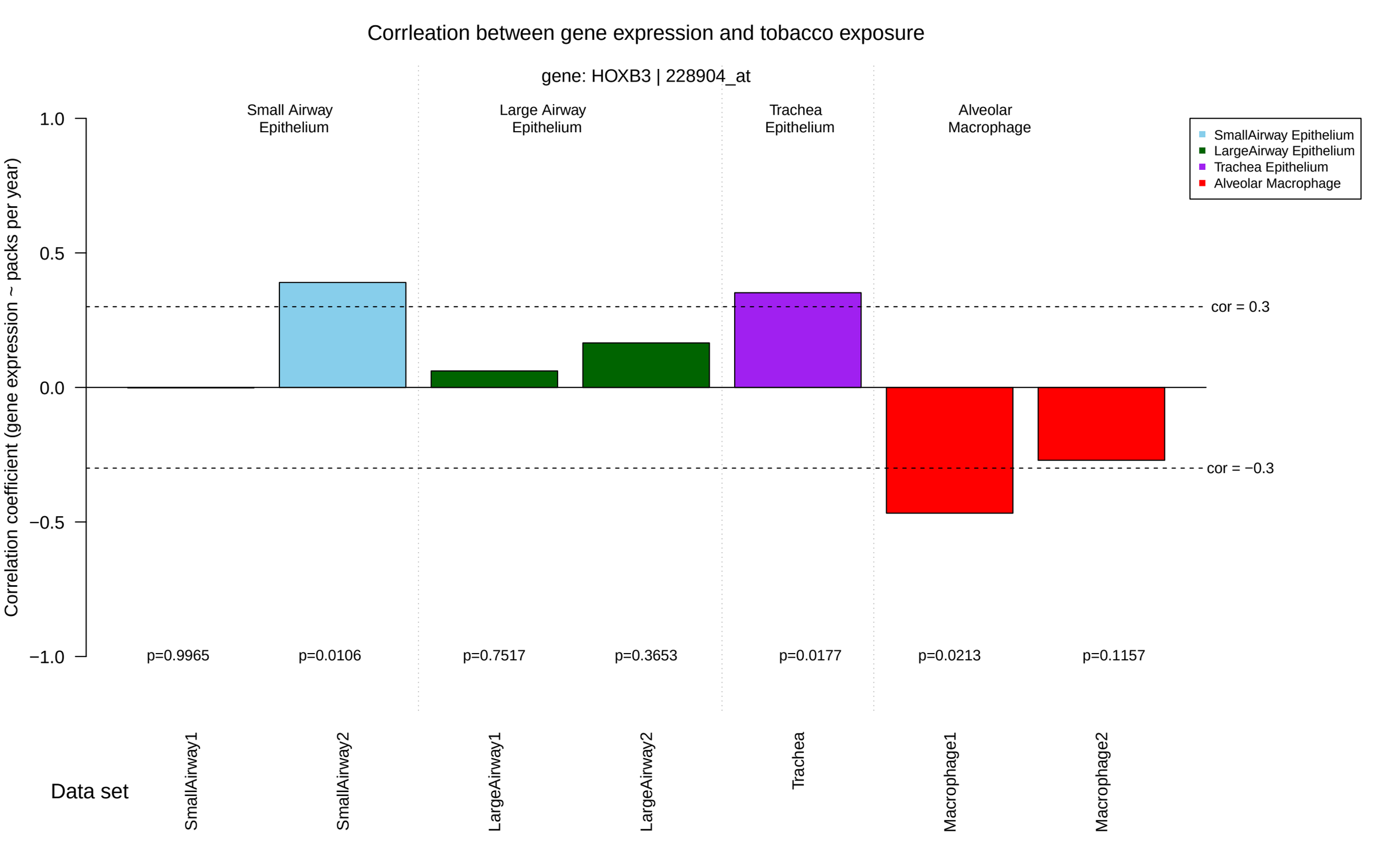

set: LargeAirway1 (p=0.2584)
HOXB3 | 228904_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.166)
HOXB3 | 228904_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
HOXB3 | 228904_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0037)
HOXB3 | 228904_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0613)
HOXB3 | 228904_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB3 | 228904_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9965
p=0.0106
p=0.7517
p=0.3653
p=0.0177
p=0.0213
p=0.1157
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2